

Example of PDF A4 page

LOGO 1

LOGO 2

cea list

FirstName LASTNAME

Affiliation 1

Affiliation 2

Affiliation 3

ORCID : 0000-0000-0000-0000

professional@address.1

professional@address.2

DOCUMENT TYPE

THE TITLE OF YOUR MANUSCRIPT

February 12, 2025

Thesis Adviser: FirstName LASTNAME

Thesis Co-Supervisors: FirstName LASTNAME

FirstName LASTNAME

CSI Members: FirstName LASTNAME (HDR)

FirstName LASTNAME

A School, A Speciality

20xx – 20xx Academic Years

From xx.xx.20xx to xx.xx.20xx

Acknowledgements

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Quote here. Quote here. Quote here. Quote here. Quote here. Quote here. Quote here.
Quote here. Quote here. Quote here.

Author, "Title".

Résumé

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Abstract

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

List of Publications

This thesis is based on the following publications:

- I Valentin Isaac--Chassande, Adrian Evans, Yves Durand, and Frédéric Rousseau. 2024. Dedicated Hardware Accelerators for Processing of Sparse Matrices and Vectors: A Survey. *ACM Trans. Archit. Code Optim.* 21, 2, Article 27 (Feb. 2024), 26 pages. Reference [5]
- II Valentin Isaac--Chassande, Adrian Evans, Yves Durand, and Frédéric Rousseau. 2024. SpDCache: Region-Based Reduction Cache for Outer-Product Sparse Matrix Kernels. In *2024 IEEE 35th International Conference on Application-specific Systems, Architectures and Processors (ASAP)*. IEEE Computer Society, Los Alamitos, CA, USA, 3–7. Reference [6]

The papers will be referred to in the thesis using their Roman numerals.

Summary

Acknowledgements	1
Epigraph	2
Résumé	3
Abstract	4
List of Publications	5
Summary	6
Notations	8
Glossary	9
Acronyms	10
1 Recommendations	11
2 A Chapter	13
2.1 A Section	13
2.2 An Other Section	13
3 An Other Chapter Title	20
3.1 Section Title	20
Conclusion	21
1 First element	21
2 Second element	22
Appendices	23
A First element	23
B Second element	24
List of Figures	25
List of Tables	26

List of Algorithms	27
List of Source Code	28
Bibliography	29
Contents	31

Notations

$$\forall (a, b) \in \mathbb{Z}^2, \llbracket a, b \rrbracket = [a, b] \cap \mathbb{Z}$$

...

Glossary

- glossary test 4** Is the glossary test entry number four plural. 17
- glossary test 3** Is the glossary test entry number three plural. 17
- glossary test 2** Is the glossary test entry number two plural. 17
- glossary test 1** Is the glossary test entry number one. 17
- latex** Is a mark up language specially suited for scientific documents. 17

Acronyms

- APTtest1** Acronym Plural Form Test Number One. 17
- APTtest2** Acronym Plural Form Test Number Two. 17
- APTtest3** Acronym Plural Form Test Number Three. 18
- APTtest4** Acronym Plural Form Test Number Four. 18
- OP** Outer-Product. 17

Chapter 1

Recommandations

Voici quelques règles et recommandations CEA pour l'écriture de votre mémoire :

1. Règles de confidentialité : Les informations que vous avez pu récolter durant votre temps au CEA peuvent être confidentielles. Merci de vous rapprocher de votre tuteur de stage pour les connaître. Apposez une mention *Confidentiel* sur la première page de votre document ainsi qu'à chaque page qui le nécessite (en-tête ou pied de page) le cas échéant. Pour rappel : voir Figure 1.1. Plus de détails sont donnés dans le [guide de protection de l'information](#).

		DIFFUSION RESTREINTE	CONFIDENTIEL + Portée		DIFFUSION LIMITEE	PUBLIC-PUBLIE
Impact sur l'information	Personnelle 	(Si atteinte aux intérêts de la nation)	Dès que risque avéré		Restriction de diffusion	Aucune confidentialité
	Dont la portée n'excède pas le CEA 	-	Préjudice moyen et fort pour le laboratoire/unité		Préjudice faible pour le laboratoire/unité	
	Concerne la PPST de la Nation 	Risques PPST fort (hors besoin de classifier)	Risques d'ingérence et perte d'avancée moyens	Risques PPST faibles	Risques PPST nuls	

Figure 1.1: Les niveaux de protection de l'information en fonction de leur sensibilité

2. Propriété intellectuelle : **Une vigilance importante** doit être apportée à **toutes informations scientifiques diffusées** dans le cadre de votre mémoire ET de votre soutenance, qui ne soient pas déjà dans le domaine public. La diffusion d'information brevetable, dans un cadre non confidentiel, peut être motif d'une demande de brevet non recevable. Si cela vous concerne, vous pouvez signer un NDA (accord de confidentialité) avec votre école. Pour la soutenance, soyez vigilant au tiers qui peuvent être présents et qui n'auraient pas signé de NDA.
3. Médiathèque : Pour vous aider dans l'illustration de votre rapport et dans la description du CEA, une médiathèque est disponible sur l'intranet : [La médiathèque - Accueil \(cea.fr\)](#)
4. Sources et bibliographies : Pour éviter toute forme de plagiat (c'est une faute grave !), vous devez citer les auteurs et les œuvres dont s'est nourrie votre réflexion.

5. Intégrité scientifique et règles : Le CEA a adhéré au code de conduite européen pour l'intégrité en recherche. Consultez le sur allea.org.
6. Recommandation d'écriture : Alignement en Justifié, annotez et recensez tous vos graphiques et schémas avec un titre et légendes, numérotez vos pages. Pour la rédaction des livrables, voir quelques conseils [ici](#), p9-10.
7. Sauvegarde et archive :
 - Pensez à sauvegarder votre mémoire en cours d'écriture sur un réseau CEA, il serait dommage de tout perdre si votre PC venait à rendre l'âme !
 - Pour la sauvegarde de la version définitive, merci de vous rapprocher de votre tuteur. Ne la laissez pas dans vos dossiers personnels du PC, qui seront supprimés à votre départ. Il en va de même pour vos travaux.

Chapter 2

A Chapter

2.1 A Section

2.1.1 Example of Text Edition

Some text. **Some text.** *Some text.* *Some text.* Some text.

Unbreakable space. Unbreakable space. Unbreakable space. Unbreakable space. Unbreakable space. Unbreakable space. Unbreakable box. Unbreakable box. Unbreakable box. Unbreakable box. Unbreakable box. Be careful with `\mbox{ }`. If you put too much text in it, $\text{\LaTeX}$ will have difficulties to space the text correctly and generate a *Warning*, as in this example.

Space before new paragraph.

Example of english quotation marks: “a quote”.

2.1.2 Example of References

In this template, we use the *natbib* package for the bibliography (see in *Configuration.tex*). You can change the bibliography style in *MAIN.tex* (`\bibliographystyle{ }`). In this template, we use a specific one imported from ACM journal templates (*ACM-Reference-Format.bst*), but there is some predefined style such as *acm*, *ieeetr*, and *plain*, the default one.

Some text with reference [10]. Some text. Some text with multiple references [1–4, 7–9].

References to publications: Paper I and Paper II.

2.2 An Other Section

2.2.1 Example of Figures

2.2.1.1 Position of the Figure

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis



Figure 2.1: Default (t): Top of the page

egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.



Figure 2.2: With h : Between the paragraphs

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

Reference to a figure: Figure 2.1.



Figure 2.3: With b : Bottom of the page

2.2.1.2 Subfigures

References: Figure 2.4, Figure 2.4a, Figure 2.4b.i.



(a) A Figure



(b.i) A Figure



(b.ii) A Figure

(b) A Subfigure with Subsubfigures

Figure 2.4: A Figure with Subfigures

2.2.1.3 Two Figures Side by Side



Figure 2.5: A Figure



Figure 2.6: A Figure

2.2.1.4 CEA Picture Library

Some “free-to-use” pictures for CEA: <https://phototheque.intra.cea.fr/>

2.2.2 Examples of Equations

In line equations with $\$. \$$: $\forall a \in \mathbb{R}, a^1 = a$

Full line equation without environment with $\$. \$ \$$:

$$\forall a \in \mathbb{R}, a^1 = a$$

Equation with environment, referenced as Equation 2.1:

$$\forall a \in \mathbb{R}, a^1 = a \quad (2.1)$$

Link for list of symbols in math mode: <https://www.cmor-faculty.rice.edu/~heinken/latex/symbols.pdf>

2.2.3 Examples of Tables

Reference to a table: Table 2.1.

	A	B
Oui	1	0
Non	0	1

Table 2.1: A Simple Table

	A	B
Oui	1	0
Non	0	1

Table 2.2: A Simple Table Using *hhline* Package

	A	B
Oui	1	0
Non	0	1
	1	0

Table 2.3: A Simple Table Using *multirow* and *multicolumn*

	A	B
Oui	1	0
Non	0	1

Table 2.4: A Simple Table with Fixed Size Columns

Table 2.5: A Complex Table with Title Above

A1	B1	C1				D1.....
A2						
A3		B2	C2		C3	
1	2	3	4	5	6	7
1	2	3	4	5	6	7
1	2	3	4	5	6	7
1	2	3	4	5	6	7

A link to an online table generator: <https://www.tablesgenerator.com/>

2.2.4 Example of Algorithms

Reference to algorithm: Algorithm 2.1

Algorithm 2.1: A Simple Algorithm with Comments

Input: N

```

1 for  $n \in \llbracket 0, N \rrbracket$  do                                /* A comment */
2    $a \leftarrow 0$ ;
3   for  $i \in \llbracket 0, 10 \rrbracket$  do
4      $rand \leftarrow \text{Random}(0, 10)$ ;                /* An other comment */
5      $a \leftarrow a + i \times rand$ ;
6   end
7   if  $a \leq 100$  then
8     return  $n$ ;
9   end
10 end

```

2.2.5 Examples of Lists

Example of list using the *itemize* environment.

- Something.
- An other thing.

Example of list using the *enumerate* environment:

1. something;
2. an other thing.

2.2.6 Use of Glossary Entries

Some text with the word Outer-Product (OP) and the word latex, and then again OP, with its plural form OPs.

Some tests of glossary commands (see source code and *Gloss&Acro.tex*) with customized glossary entries:

- glossary test 1 (*first call*) / glossary test 1 / Glossary test 1 / glossary test 1 in plural form / Glossary test 1 in plural form
- First call of gl2 in plural form (*first call*) / glossary test 2 / Glossary test 2 / glossary test 2s / Glossary test 2s
- Glossary test 3 (*first call*) / glossary test 3 / Glossary test 3 / glossary test 3s / Glossary test 3s
- First call of gl4 in plural form (*first call*) / glossary test 4 / Glossary test 4 / glossary test 4s / Glossary test 4s

Some tests with customized acronym entries (see source code and *Gloss&Acro.tex*):

- Acronym Plural Form Test Number One (APTest1) (*first call*) / APTest1 / APTest1 / APTest1s / APTest1s
- Acronym Plural Form Test Number Twos (APTest2s) (*first call*) / APTest2 / APTest2 / APTest2s / APTest2s

- Acronym Plural Form Test Number Three (APTest3) (*first call*) / APTest3 / APTest3 / APTest3s / APTest3s
- Acronym Plural Form Test Number Fours (APTest4s) (*first call*) / APTest4 / APTest4 / APTest4s / APTest4s / APTest4s

This package is flexible and has many more capabilities. See <https://tug.ctan.org/macros/latex/contrib/glossaries/glossariesbegin.pdf> for more details.

2.2.7 Some Symbols with *pifont*

The *pifont* package gives you access to some symbols that the L^AT_EX math environment do not support. For example, `\ding{51}` prints ✓.

A table of the *pifont* symbols and their matching number is available [here](#).

2.2.8 Example of Source Code

To generate the Source Code 2.1, we use the *listings* package. You need to import a source code file like *op_spmv.c* in the example, and use the custom command:

`\sourcecode{file}{title}{label}{position}`

To change the rendering of the code, you can modify the *listings* configuration parameters in *Configuration.tex*.

Source Code 2.1: A Source Code Example

```
1 startRedEngine();
2 for (int n=0; n<N; n++) { /* Column first */
3     int bn = b[n];
4     int currPos = A.ptr[n];
5     int nextPos = A.ptr[n+1];
6     for (int pos=currPos; pos<nextPos; pos++) {
7         int idx = A.idx[pos];
8         int val = A.val[pos] * bn;
9         /* Without RED engine: c[idx] += val; */
10        /* With RED engine: */
11        int *addr = c + idx;
12        region_t reg = regionSelect();
13        asm volatile ( /* RISC-V ISA: Type R */
14            "reg x0, x0, %[rs2]"
15            :
16            : [rs2] "r" ((int) reg)
17        );
18        asm volatile ( /* RISC-V ISA: Type S */
19            "red.w %[rs2], 0(%[rs1])"
20            :
21            : [rs2] "r" (addr), [rs1] "r" (val)
22        );
23    }
24 }
25 flushAndStopRedEngine();
```

2.2.9 Box Examples to Highlight Important Text

Something important!

:)

Something dangerous!

Warning

Something dangerous with a title!

Conclusion

Feel free to define your own boxes! Take a look at the *tcolorbox* package documentation for complex boxes.

Chapter 3

An Other Chapter Title

Contents

3.1	Section Title	20
3.1.1	Subsection Title	20

3.1 Section Title

3.1.1 Subsection Title

3.1.1.1 Subsubsection Title

Paragraph Title Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Conclusion

Contents

1	First element	21
1.1	A Section	21
2	Second element	22

1 First element

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

1.1 A Section

1.1.1 A Subsection

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend

consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

2 Second element

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetur adipiscing elit. In hac habitasse platea dictumst. Integer tempus convallis augue. Etiam facilisis. Nunc elementum fermentum wisi. Aenean placerat. Ut imperdiet, enim sed gravida sollicitudin, felis odio placerat quam, ac pulvinar elit purus eget enim. Nunc vitae tortor. Proin tempus nibh sit amet nisl. Vivamus quis tortor vitae risus porta vehicula.

Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam. Aliquam pellentesque, augue quis sagittis posuere, turpis lacus congue quam, in hendrerit risus eros eget felis. Maecenas eget erat in sapien mattis porttitor. Vestibulum porttitor. Nulla facilisi. Sed a turpis eu lacus commodo facilisis. Morbi fringilla, wisi in dignissim interdum, justo lectus sagittis dui, et vehicula libero dui cursus dui. Mauris tempor ligula sed lacus. Duis cursus enim ut augue. Cras ac magna. Cras nulla. Nulla egestas. Curabitur a leo. Quisque egestas wisi eget nunc. Nam feugiat lacus vel est. Curabitur consectetur.

Appendices

Contents

A	First element	23
A.1	A Section	23
B	Second element	24

A First element

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

A.1 A Section

A.1.1 A Subsection

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend

consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

B Second element

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetur adipiscing elit. In hac habitasse platea dictumst. Integer tempus convallis augue. Etiam facilisis. Nunc elementum fermentum wisi. Aenean placerat. Ut imperdiet, enim sed gravida sollicitudin, felis odio placerat quam, ac pulvinar elit purus eget enim. Nunc vitae tortor. Proin tempus nibh sit amet nisl. Vivamus quis tortor vitae risus porta vehicula.

Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam. Aliquam pellentesque, augue quis sagittis posuere, turpis lacus congue quam, in hendrerit risus eros eget felis. Maecenas eget erat in sapien mattis porttitor. Vestibulum porttitor. Nulla facilisi. Sed a turpis eu lacus commodo facilisis. Morbi fringilla, wisi in dignissim interdum, justo lectus sagittis dui, et vehicula libero dui cursus dui. Mauris tempor ligula sed lacus. Duis cursus enim ut augue. Cras ac magna. Cras nulla. Nulla egestas. Curabitur a leo. Quisque egestas wisi eget nunc. Nam feugiat lacus vel est. Curabitur consectetur.

List of Figures

1.1	Les niveaux de protection de l'information en fonction de leur sensibilité .	11
2.1	Default (<i>t</i>): Top of the page	14
2.2	With <i>h</i> : Between the paragraphs	14
2.3	With <i>b</i> : Bottom of the page	14
2.4	A Figure with Subfigures	15
2.5	A Figure	15
2.6	A Figure	15

List of Tables

2.1	A Simple Table	16
2.2	A Simple Table Using <i>hhline</i> Package	16
2.3	A Simple Table Using <i>multirow</i> and <i>multicolumn</i>	16
2.4	A Simple Table with Fixed Size Columns	16
2.5	A Complex Table with Title Above	16

List of Algorithms

2.1 A Simple Algorithm with Comments	17
--	----

List of Source Code

2.1 A Source Code Example	18
-------------------------------------	----

Bibliography

- [1] Richard Barrett, Michael Berry, Tony F. Chan, James Demmel, June Donato, Jack Dongarra, Victor Eijkhout, Roldan Pozo, Charles Romine, and Henk van der Vorst. 1994. *Templates for the Solution of Linear Systems: Building Blocks for Iterative Methods*. Society for Industrial and Applied Mathematics. <https://doi.org/10.1137/1.9781611971538>
- [2] Scott Beamer, Krste Asanović, and David Patterson. 2017. The GAP Benchmark Suite. arXiv:1508.03619 [cs.DC] <http://gap.cs.berkeley.edu/benchmark.html>
- [3] Achi Brandt and Dorit Ron. 2003. *Multigrid Solvers and Multilevel Optimization Strategies*. Springer US, Boston, MA, 1–69. https://doi.org/10.1007/978-1-4757-3748-6_1
- [4] Mehmet Deveci, Simon David Hammond, Michael M. Wolf, and Sivasankaran Rajamanickam. 2018. Sparse Matrix-Matrix Multiplication on Multilevel Memory Architectures: Algorithms and Experiments. (4 2018). <https://doi.org/10.2172/1435688>
- [5] Valentin Isaac--Chassande, Adrian Evans, Yves Durand, and Frédéric Rousseau. 2024. Dedicated Hardware Accelerators for Processing of Sparse Matrices and Vectors: A Survey. *ACM Trans. Archit. Code Optim.* 21, 2, Article 27 (Feb. 2024), 26 pages. <https://doi.org/10.1145/3640542>
- [6] Valentin Isaac--Chassande, Adrian Evans, Yves Durand, and Frédéric Rousseau. 2024. SpDCache: Region-Based Reduction Cache for Outer-Product Sparse Matrix Kernels. In *2024 IEEE 35th International Conference on Application-specific Systems, Architectures and Processors (ASAP)*. IEEE Computer Society, Los Alamitos, CA, USA, 3–7. <https://doi.org/10.1109/ASAP61560.2024.00012>
- [7] JESD235A 2016. *JEDEC Updates Groundbreaking High Bandwidth Memory (HBM) Standard*. Technical Report. JEDEC. <https://www.jedec.org/news/pressreleases/jedec-updates-groundbreaking-high-bandwidth-memory-hbm-standard>
- [8] Shaden Smith, Jee W. Choi, Jiajia Li, Richard Vuduc, Jongsoo Park, Xing Liu, and George Karypis. 2017. *FROSTT: The Formidable Repository of Open Sparse Tensors and Tools*. <http://frostdt.io/>

- [9] Sriseshan Srikanth, Thomas M. Conte, Erik P. DeBenedictis, and Jeanine Cook. 2017. The superstrider architecture: Integrating logic and memory towards non-von Neumann computing. *2017 IEEE International Conference on Rebooting Computing, ICRC 2017 - Proceedings* 2017-January (2017), 1 – 8. <https://doi.org/10.1109/ICRC.2017.8123669>
- [10] Guowei Zhang, Nithya Attaluri, Joel S. Emer, and Daniel Sanchez. 2021. Gamma: Leveraging Gustavson’s Algorithm to Accelerate Sparse Matrix Multiplication. In *Proceedings of the 26th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (Virtual, USA) (ASPLOS ’21)*. Association for Computing Machinery, New York, NY, USA, 687–701. <https://doi.org/10.1145/3445814.3446702>

Contents

Acknowledgements	1
Epigraph	2
Résumé	3
Abstract	4
List of Publications	5
Summary	6
Notations	8
Glossary	9
Acronyms	10
1 Recommendations	11
2 A Chapter	13
2.1 A Section	13
2.1.1 Example of Text Edition	13
2.1.2 Example of References	13
2.2 An Other Section	13
2.2.1 Example of Figures	13
2.2.1.1 Position of the Figure	13
2.2.1.2 Subfigures	15
2.2.1.3 Two Figures Side by Side	15
2.2.1.4 CEA Picture Library	15
2.2.2 Examples of Equations	15
2.2.3 Examples of Tables	16
2.2.4 Example of Algorithms	17
2.2.5 Examples of Lists	17
2.2.6 Use of Glossary Entries	17
2.2.7 Some Symbols with <i>pifont</i>	18
2.2.8 Example of Source Code	18

2.2.9	Box Examples to Highlight Important Text	19
3	An Other Chapter Title	20
3.1	Section Title	20
3.1.1	Subsection Title	20
3.1.1.1	Subsubsection Title	20
	Paragraph Title	20
	Conclusion	21
1	First element	21
1.1	A Section	21
1.1.1	A Subsection	21
2	Second element	22
	Appendices	23
A	First element	23
A.1	A Section	23
A.1.1	A Subsection	23
B	Second element	24
	List of Figures	25
	List of Tables	26
	List of Algorithms	27
	List of Source Code	28
	Bibliography	29
	Contents	31